SFWRENG 3K04: Software Development

Deliverable 1: Requirements Document

Hospital Monitoring System

Team [35], Lab Section [L03]
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1. Introduction And System Contents

The Hospital Monitoring System (HMS) is a centralized supervisory application that communicates with multiple Automated Dispensing Cabinets (ADCs) deployed throughout a hospital. It provides authorized hospital staff with information about medication inventory, alarms, cabinet health, operational status, and historical activity.

The HMS supports ADC communication, inventory supervision, alarm management, historical data recording and review, report generation, and an operator dashboard. Users include pharmacists, nurses, technicians, and hospital administrators, with access restricted by assigned roles.

The ADC system manages local dispensing operations and directly controls cabinet hardware, including locks, sensors, and actuators. The HMS receives information from the ADCs and exchanges supervisory requests and configuration information through the agreed communication interface.

This document defines the HMS scope, functional and non-functional requirements, user stories and acceptance criteria, high-level architecture, and structural and behavioural models. It also records requirement traceability and the assumptions and decisions used to define the system.

2. Functional Requirements

The Hospital Monitoring System (HMS) shall support communication with Automated Dispensing Cabinets (ADCs), inventory supervision, alarm management, historical data recording, report generation, an operator dashboard, and cabinet-health monitoring.

An authorized operator is a user whose assigned role permits the requested operation. Configured timing limits, inventory thresholds, status boundaries, and conflict-detection rules referenced below shall be documented before acceptance testing.

A. ADC Communication

Identifier	Requirement
HMS-ADC-00	The HMS shall maintain bidirectional communication with each connected ADC through the agreed communication interface.
HMS-ADC-01	The HMS shall uniquely identify each connected ADC using a cabinet_id.
HMS-ADC-02	The HMS shall verify communication with each connected ADC by sending heartbeat requests at the configured heartbeat interval.

HMS-ADC-03	Each heartbeat request shall contain a randomly generated mathematical challenge. The HMS shall accept the correct response to that challenge as its acknowledgement.
HMS-ADC-04	The HMS shall classify an incorrect heartbeat response, or failure to receive a response within the configured heartbeat timeout, as a communication error for the affected ADC.
HMS-ADC-05	The HMS shall receive medication-dispensing, restocking, inventory, cabinet-status, sensor-status, and maintenance information from each ADC.
HMS-ADC-06	The HMS shall associate each received ADC message with its originating cabinet_id and a receipt timestamp.
HMS-ADC-07	The HMS shall generate a stale-data alert when ADC information exceeds the configured freshness limit or contains a timestamp outside the permitted timestamp tolerance. The alert shall identify the affected ADC and information category.

B. Inventory Supervision

For these requirements, an inventory location is identified by its cabinet, drawer, and pocket identifiers. Inventory percentage is the recorded quantity divided by the location's configured full-stock quantity, multiplied by 100%.

Identifier	Requirement
HMS-IS-00	The HMS shall uniquely identify each medication product using a medication_id and associate it with its Drug Identification Number (DIN), name, strength, and dosage form.
HMS-IS-01	The HMS shall support the same medication product being stored in multiple ADCs and in multiple inventory locations within an ADC.
HMS-IS-02	The HMS shall identify each inventory location using at least cabinet_id, drawer_id, and pocket_id, and associate that location with its stored medication_id.
HMS-IS-03	The HMS shall independently track the recorded medication quantity at each inventory location.
HMS-IS-04	The HMS shall retain the batch or lot number associated with medication inventory when that information is supplied.

HMS-IS-05	The HMS shall display the recorded medication quantity and calculate its percentage of the configured full-stock quantity for each inventory location. If the full-stock quantity is unavailable or invalid, the percentage shall be displayed as unavailable.
HMS-IS-06	The HMS shall classify inventory as Full, Half, Low, or Empty using documented, non-overlapping quantity or percentage ranges that cover all valid inventory quantities. A quantity of zero shall be classified as Empty.
HMS-IS-07	The HMS shall allow authorized operators to configure a restock threshold for the medication at each inventory location, expressed as either an absolute quantity or a percentage of the configured full-stock quantity.
HMS-IS-08	The HMS shall generate a configuration alert identifying any active inventory location that lacks its required restock threshold.
HMS-IS-09	The HMS shall generate a restock alert when the recorded quantity or percentage at an inventory location reaches or falls below its configured restock threshold.
HMS-IS-10	When the recorded stock at a requested location is insufficient, the HMS shall identify alternate locations containing the same medication product and display their recorded quantities.
HMS-IS-11	The HMS shall allow authorized operators to report a discrepancy between the observed physical inventory and the quantity recorded for an inventory location.
HMS-IS-12	The HMS shall generate an inventory-discrepancy alarm upon accepting a discrepancy report, identifying the affected inventory location.

C. Alarm Management

Alarm acknowledgement means that an authorized user has recognized an alarm. Resolution means that the underlying alarm condition has been confirmed cleared.

Identifier	Requirement
HMS-AM-00	The HMS shall generate an inventory-discrepancy alarm when a discrepancy is reported by an ADC or an authorized operator.
HMS-AM-01	The HMS shall generate a device-failure alarm when an ADC reports a hardware or sensor failure.

HMS-AM-02	The HMS shall generate a communication alarm when communication with an ADC fails, including an incorrect heartbeat response or a response timeout.
HMS-AM-03	The HMS shall generate a security alarm after three consecutive denied authorization attempts within the configured counting scope. The scope and counter-reset conditions shall be documented.
HMS-AM-04	The HMS shall generate an unsafe-state alarm when an ADC reports an unsafe condition, including a cabinet door or safety interlock remaining open beyond its permitted operating condition or duration.
HMS-AM-05	The HMS shall generate a stale-data alarm when ADC information or status timestamps fail the freshness criteria defined for HMS-ADC-07.
HMS-AM-06	The HMS shall detect conflicting patient medication schedules or medication data using documented conflict-detection rules and generate an alarm identifying the affected records and conflict type.
HMS-AM-07	The HMS shall allow authorized operators to manually report an observed ADC fault and generate an alarm identifying the affected ADC and reported problem.
HMS-AM-08	Each alarm record shall contain its alarm type, source, detection timestamp, current lifecycle state, and information identifying the condition that triggered it.
HMS-AM-09	The HMS shall continue to indicate an alarm while its underlying condition remains unresolved. Acknowledgement may change the indication to acknowledged, but shall not remove its active status or indicate that the fault has been resolved.

D. Recorded Data

Identifier	Requirement
HMS-RD-00	The HMS shall record every medication-dispensing transaction reported by an ADC.
HMS-RD-01	The HMS shall record every medication-restocking transaction reported by an ADC.
HMS-RD-02	Each dispense or restock record shall include the medication identifier, quantity, cabinet and inventory-location identifiers, operator identifier, and event timestamp. Batch or lot information shall be retained when supplied, and patient identifiers shall be retained for patient-associated transactions when supplied. Unavailable optional information shall be explicitly identified.

HMS-RD-03	The HMS shall retain an inventory history linking recorded quantity changes to the corresponding transactions at each ADC inventory location.
HMS-RD-04	The HMS shall record system alarms and alerts, including their type, source, timestamp, and lifecycle state.
HMS-RD-05	The HMS shall maintain a maintenance history for each ADC.
HMS-RD-06	Each maintenance record shall identify the affected ADC, maintenance activity, responsible operator or technician, and event timestamp.
HMS-RD-07	The HMS shall record rejected updates, including patient medication schedule and medication-data updates, with the rejection reason and timestamp.
HMS-RD-08	The HMS shall record authorization failures, including the timestamp, source, and attempted operation, and identify when the repeated-denial condition in HMS-AM-03 is reached.
HMS-RD-09	The HMS shall retain the inventory baseline and transaction information needed to reconstruct recorded inventory and ADC activity at a requested time within the retained history. If missing information prevents reconstruction, the HMS shall identify the result as incomplete.

E. Report Generation

Identifier	Requirement
HMS-RG-00	The HMS shall generate medication-dispensing reports from recorded dispensing transactions.
HMS-RG-01	The HMS shall generate medication-restocking reports from recorded restocking transactions.
HMS-RG-02	The HMS shall generate ADC maintenance reports from recorded maintenance activities.
HMS-RG-03	The HMS shall generate reports of rejected patient medication schedules, medication-data updates, and other recorded rejected updates.
HMS-RG-04	The HMS shall generate reports of recorded system alarms and alerts.

HMS-RG-05	Each report shall include event timestamps and source identifiers. Transaction reports shall include medication identifiers, quantities, locations, and operator identifiers; maintenance reports shall include maintenance activities and responsible personnel; rejected-update reports shall include rejection reasons; alarm reports shall include alarm types and lifecycle states.
HMS-RG-06	The HMS shall support daily and weekly reports and identify the start, end, and time zone of the reporting period used.

F. Operator Dashboard

Identifier	Requirement
HMS-DB-00	The HMS dashboard shall display the latest received operational status of each connected ADC within the configured dashboard-update limit.
HMS-DB-01	The dashboard shall display the latest recorded medication quantities and inventory classifications for each ADC inventory location.
HMS-DB-02	The dashboard shall display a visible pop-up notification when a new alert or alarm is generated.
HMS-DB-03	Critical alarms shall remain visually prominent while unresolved. Their presentation shall distinguish unacknowledged alarms from acknowledged alarms without implying that acknowledgement has cleared the fault.
HMS-DB-04	The dashboard shall provide access to an incident log containing ADC alarms, alerts, faults, and their timestamps.
HMS-DB-05	The dashboard shall display ADC communication status and cabinet-health status separately, identifying stale or unavailable information.
HMS-DB-06	The dashboard shall display Full, Half, Low, or Empty inventory classifications using the same rules defined in HMS-IS-06.

G. Cabinet-Health Monitoring

Identifier	Requirement
HMS-CHM-00	The HMS shall receive ADC health and status updates at the reporting interval agreed with the ADC sub-team and identify updates that exceed the configured freshness limit.
HMS-CHM-01	The HMS shall monitor the main power-supply status reported by each ADC.

HMS-CHM-02	Where an ADC has an emergency or backup power supply, the HMS shall monitor its reported availability and status.
HMS-CHM-03	Where an ADC has active cooling, the HMS shall monitor its reported cooling-system status.
HMS-CHM-04	The HMS shall monitor and record the system uptime reported by each ADC.
HMS-CHM-05	The HMS shall monitor the operational status of hardware components reported by each ADC.
HMS-CHM-06	The HMS shall monitor the operational status of sensors reported by each ADC.
HMS-CHM-07	The HMS shall calculate the elapsed time since each ADC's most recent recorded maintenance activity. If no maintenance record exists, that condition shall be displayed.
HMS-CHM-08	The HMS shall generate a device-failure alarm when reported hardware or sensor health indicates a failure. A successful heartbeat shall not independently clear that failure indication.
HMS-CHM-09	Each ADC health update shall include a source timestamp. The HMS shall use the timestamp and expected reporting interval to identify stale or missing health information and shall not present that information as confirmed current.

3. Non-functional requirements

The Hospital Monitoring System (HMS) shall satisfy the following safety, security, reliability, availability, performance, traceability, and usability requirements.

These requirements apply to the HMS. Direct control of dispensing hardware and enforcement of local cabinet safety remain responsibilities of the Automated Dispensing Cabinet (ADC).

Category	Identifier	Requirement
Safety	SAF-00	The HMS shall prevent state-changing supervisory requests from being sent to an Automated Dispensing Cabinet (ADC) when communication with that ADC is unavailable or has not been verified.

Safety	SAF-01	The HMS shall identify stale or unavailable ADC information rather than present it as confirmed current information.
Safety	SAF-02	Acknowledging an alarm shall not clear its underlying fault indication or independently change the ADC's operational status.
Security	SEC-00	The HMS shall restrict protected functions and records to authenticated users.
Security	SEC-01	The HMS shall record authentication failures and denied attempts to access protected functions, including the timestamp and attempted operation.
Security	SEC-02	The HMS shall enforce Role-Based Access Control (RBAC) according to a documented role-permission matrix.
Security	SEC-03	Communication carrying protected information shall use the agreed security mechanism to authenticate endpoints and protect information against unauthorized disclosure or modification.
Reliability	REL-00	Loss of communication with one ADC shall not prevent the HMS from monitoring other connected ADCs.
Reliability	REL-01	Invalid ADC messages shall be rejected without replacing previously accepted information or terminating monitoring of other ADCs.
Reliability	REL-02	The HMS shall retain successfully saved configuration, transaction, and alarm records across a normal application shutdown and restart.
Reliability	REL-03	Receiving the same ADC event more than once shall not create duplicate transactions or apply the same inventory change more than once.
Availability	AVA-00	During an ADC disconnection, the HMS shall continue to provide access to retained records and clearly identify affected live information as unavailable or stale.
Availability	AVA-01	Following reconnection, the HMS shall resume monitoring the ADC and mark information as current only after receiving a valid fresh update.
Performance	PER-00	Under the documented test workload, the HMS shall display the result or failure of an ordinary dashboard

		query within the agreed user-response limit.
Performance	PER-01	The HMS shall detect communication loss within the configured heartbeat timeout measured from transmission of the unanswered heartbeat request.
Performance	PER-02	The HMS shall display accepted ADC updates and generated alarms within their respective agreed display-update limits.
Traceability / Auditability	TRA-00	The HMS shall retain records of medication transactions, alarms, alarm acknowledgements, and changes to inventory thresholds.
Traceability / Auditability	TRA-01	Each recorded event shall include its event type, timestamp, affected ADC or inventory location where applicable, and initiating user or system identifier.
Traceability / Auditability	TRA-02	The HMS shall retain the event identifiers and ordering information needed to trace recorded inventory changes and shall identify known gaps that prevent complete historical reconstruction.
Usability / Accessibility	USA-00	The HMS shall use consistent ADC identifiers, inventory-location identifiers, status labels, and quantity units across its dashboard, records, and reports.
Usability / Accessibility	USA-01	Alarm severity and status shall be communicated through text or symbols in addition to colour, without relying solely on blinking or sound.
Usability / Accessibility	USA-02	The dashboard shall distinguish communication status, cabinet health, operational status, and information freshness.
Usability / Accessibility	USA-03	Core workflows for viewing inventory, reviewing and acknowledging alarms, and requesting reports shall be operable using a keyboard with a visible focus indicator.

4. User Stories

The following user stories describe the primary needs of stakeholders interacting with the HMS. Each story identifies a specific stakeholder, their goal, and the value provided by achieving that goal. Stories are grouped according to the functional capabilities of the HMS and linked to the corresponding system requirements. Acceptance criteria are expressed using Given-When-Then scenarios to define observable conditions under which each story is considered successfully implemented.

Inventory Supervision

User Story:

As a pharmacist, I want to see where medication is stored and how much remains at each ADC inventory location so that I can identify locations that need restocking and find alternate stock when needed.

Scenario 1: Check Medication Availability	
Acceptance Criteria	**Description**
Given	The same medication is stored across multiple ADCs or inventory locations
When	The pharmacist checks the medication's inventory
Then	The HMS displays the quantity and inventory state for each location containing that medication

Scenario 2: Restock Threshold Reached	
Acceptance Criteria	**Description**
Given	A medication location has a configured restock threshold
When	The inventory reaches or falls below that threshold
Then	The HMS generates a restock alert identifying the affected medication and inventory location

Scenario 3: Physical Stock does not match	
Acceptance Criteria	**Description**
Given	An authorized operator observes that the physical medication quantity does not match the quantity recorded by the HMS
When	The operator reports the discrepancy
Then	The HMS generates an inventory discrepancy alarm for the affected inventory location

Alarm Management

User Story:
As a pharmacist, I want alarms to clearly identify what went wrong and which ADC is affected so that I can determine what requires attention without having to inspect every cabinet.

Scenario 1: ADC Fault Detected	
Acceptance Criteria	**Description**
GIVEN	An ADC reports a hardware or sensor failure
WHEN	The HMS receives the failure information
THEN	The HMS generates an alarm associated with the affected ADC

Scenario 2: Alarm Details Recorded	
Acceptance Criteria	**Description**
GIVEN	The HMS generates an alarm
WHEN	The alarm is recorded by the HMS
THEN	The HMS stores the alarm type, originating device, timestamp, current state, and associated information.

Scenario 3: Unresolved Critical Alarm	
Acceptance Criteria	**Description**
GIVEN	An active alarm condition has not been acknowledged or resolved
WHEN	The pharmacist continues monitoring the HMS
THEN	The HMS continues to indicate that the alarm is active

ADC Communication

User Story: As a technician, I want the HMS to identify when an ADC stops responding or begins providing stale information so that I can distinguish a communication problem from a healthy cabinet.

Scenario 1: Successful Heartbeat	
Acceptance Criteria	**Description**
Given	An ADC is connected and communicating with the HMS
When	The ADC correctly responds to the HMS heartbeat challenge
Then	The HMS continues to identify communication with that ADC as operational

Scenario 2: Heartbeat Failure	
Acceptance Criteria	**Description**
Given	The HMS sends a heartbeat challenge to a connected ADC

When	The ADC returns an incorrect response or does not respond within the configured timeout
Then	The HMS identifies a communication error for that ADC

Cabinet Health Monitoring

User Story: As a nurse, I want to view the operational health and status of each ADC so that I can determine whether a cabinet is available for reliable use and identify when a cabinet requires attention.

Scenario 1: Check Cabinet Health	
Acceptance Criteria	**Description**
GIVEN	Current health information has been received from an ADC
WHEN	The nurse checks the cabinet's health status
THEN	The HMS displays the available power, cooling, hardware, sensor, and operational status of the ADC

Scenario 2: Cabinet Health Failure	
Acceptance Criteria	**Description**
GIVEN	An ADC reports a hardware or sensor failure
WHEN	The HMS processes the updated health information
THEN	The HMS identifies the affected ADC as requiring attention and generates the appropriate device-failure alarm

Scenario 3: Cabinet Becomes Unavailable

Acceptance Criteria	Description
GIVEN	An ADC was previously operating normally
WHEN	Its reported health information indicates that it is no longer operational
THEN	The HMS identifies the affected ADC as unavailable and generates the appropriate device-failure alarm.

Recorded Data

User Story: As a hospital administrator, I want to review a complete record of medication transactions and ADC events so that I can verify system activity, investigate discrepancies, and maintain accountability for actions performed through the HMS.

Scenario 1: Trace Medical Transaction	
Acceptance Criteria	**Description**
GIVEN	A medication dispense or restock transaction has occurred at an ADC
WHEN	The administrator reviews the recorded transaction
THEN	The HMS displays the applicable medication, quantity, ADC location, operator, patient, and timestamp information

Scenario 2: Investigate Discrepancies in Inventory	
Acceptance Criteria	**Description**
GIVEN	A discrepancy exists between recorded and expected medication inventory
WHEN	The administrator reviews the transaction history for the affected inventory location

THEN	The HMS provides the recorded transactions needed to trace changes in medication quantity at that location

Scenario 3: Review Significant System Events	
Acceptance Criteria	**Description**
GIVEN	Alarms, authorization failures, rejected updates, or maintenance activities have been recorded
WHEN	The administrator reviews the historical activity of an ADC
THEN	The HMS displays the recorded events with their associated ADC and event information

Report Generation

As an Administrator, I want to generate reports summarizing medication transactions and ADC activity so that I can evaluate medication operations and identify recurring issues that require attention.

Scenario 1: Review Medication Activity	
Acceptance Criteria	**Description**
GIVEN	Medication dispensing and restocking transactions have been recorded
WHEN	The Administrator requests a medication activity report
THEN	The HMS generates a report containing the relevant dispensing and restocking transactions

Scenario 2: Review Operational Events	
Acceptance Criteria	**Description**
GIVEN	Maintenance activities, rejected updates, or alarms have been recorded
WHEN	The Administrator requests the corresponding operational report
THEN	The HMS generates a report containing the relevant events, timestamps, and associated ADC identifiers

Scenario 3: Review Activity Over Time	
Acceptance Criteria	**Description**
GIVEN	HMS activity has been recorded over multiple days
WHEN	The Administrator requests a daily or weekly report
THEN	The HMS generates the report using information from the requested reporting period

Dashboard

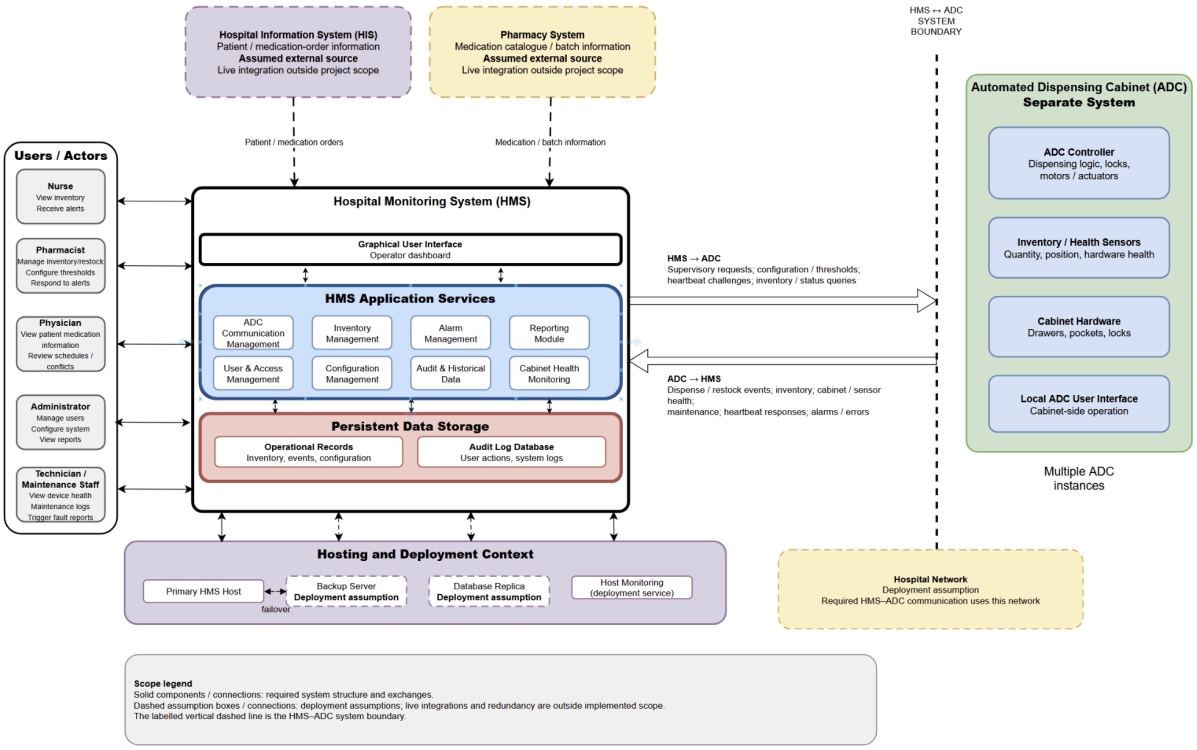
User Story: As a pharmacist, I want a centralized dashboard that brings together ADC status, medication inventory, and active alarms so that I can quickly determine which cabinets require attention during normal hospital operation.

Scenario 1: Monitor ADC Status	
Acceptance Criteria	**Description**
GIVEN	Multiple ADCs are connected to the HMS
WHEN	The pharmacist accesses the dashboard
THEN	The HMS displays the current operational, communication, and health status of the connected ADCs

Scenario 2: Monitor Medication Inventory	
Acceptance Criteria	**Description**
GIVEN	Current medication inventory information has been received from the ADCs
WHEN	The pharmacist views the dashboard
THEN	The HMS displays current inventory quantities and identifies inventory states as Full, Half, Low, or Empty

Scenario 3: Critical Alarm Notification	
Acceptance Criteria	**Description**
GIVEN	The HMS generates a critical alarm associated with an ADC
WHEN	The alarm is presented on the dashboard
THEN	The HMS displays a visible notification and keeps the critical alarm visually prominent while unresolved, distinguishing acknowledged from unacknowledged alarms

5. High-level architecture diagram



High-Level Architecture Diagram Description

The high-level architecture diagram shows the major components of the Hospital Monitoring System (HMS), its users, and its communication boundary with the Automated Dispensing Cabinet (ADC) system. The HMS contains an operator interface, application services, and persistent data storage. These components support inventory supervision, alarm management, cabinet-health monitoring, historical records, reporting, user access, and configuration.

Hospital staff interact with the HMS through the operator interface. The HMS exchanges monitoring information and agreed supervisory requests with ADCs through its communication component. Each ADC remains responsible for local dispensing operations, hardware control, and local safety enforcement.

The Hospital Information System and Pharmacy System are shown as assumed external information sources. Their connections represent the intended deployment context rather than a commitment to implement live integrations in this project. Where a specified HMS function requires their information, an agreed interface or documented simulation shall provide the necessary inputs.

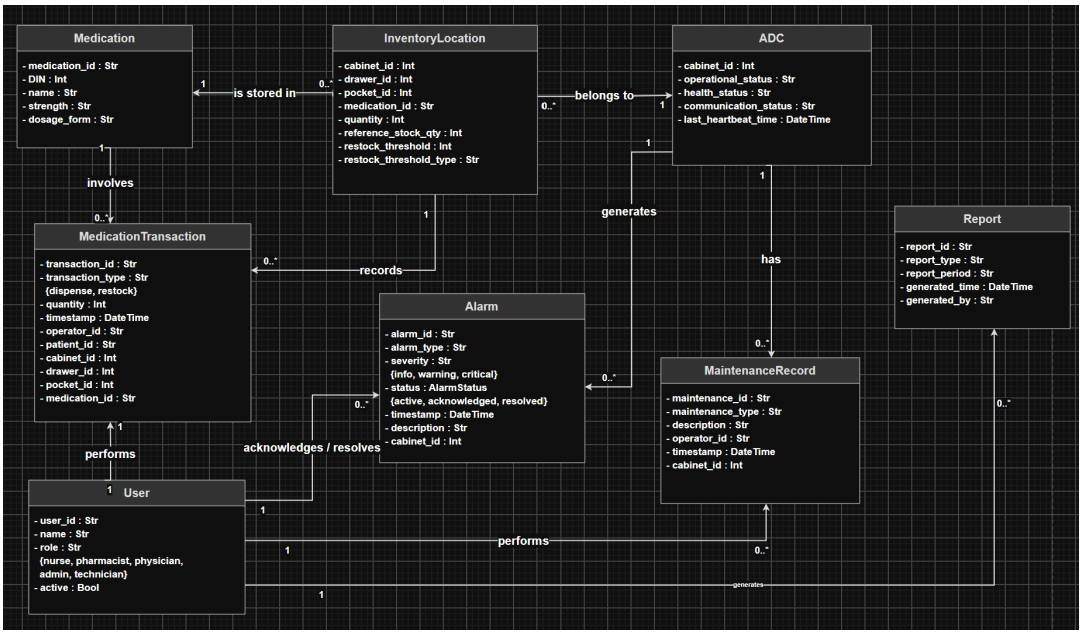
Backup servers and database replication are also deployment assumptions. They illustrate infrastructure that may support a hospital deployment but are not additional implementation or demonstration requirements for this deliverable. The HMS remains responsible for the persistence and communication-recovery behavior explicitly stated in its requirements.

The diagram distinguishes required HMS components from assumed deployment infrastructure and external integrations. Detailed protocols, replication mechanisms, and failover designs are outside the scope of this high-level architecture.

6. UML models — structure and behaviour

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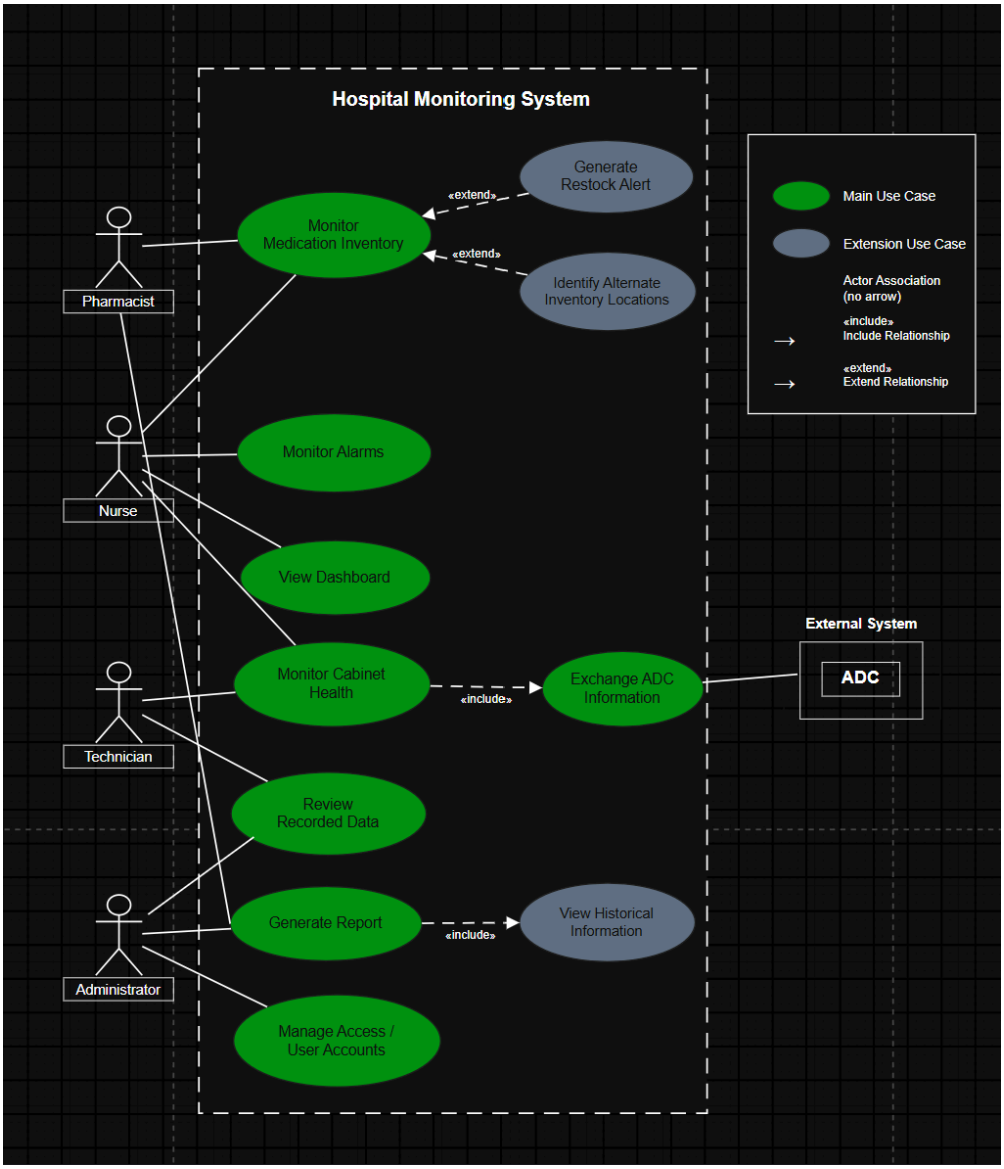
a. Conceptual domain / class model (Structure)



The following conceptual domain model illustrates the elements of the Hospital Monitoring System (HMS) and their relationships. The model shows how the system represents medication information, inventory locations, transactions, ADC operational status, maintenance records, alarms, reports, and users. These relationships provide a structural overview of how the HMS organizes and

manages information received from ADCs, while maintaining the separation between monitoring and the physical dispensing operations performed by individual ADCs.

b. Use case diagram(s) (Behaviour):



Use Case Diagram

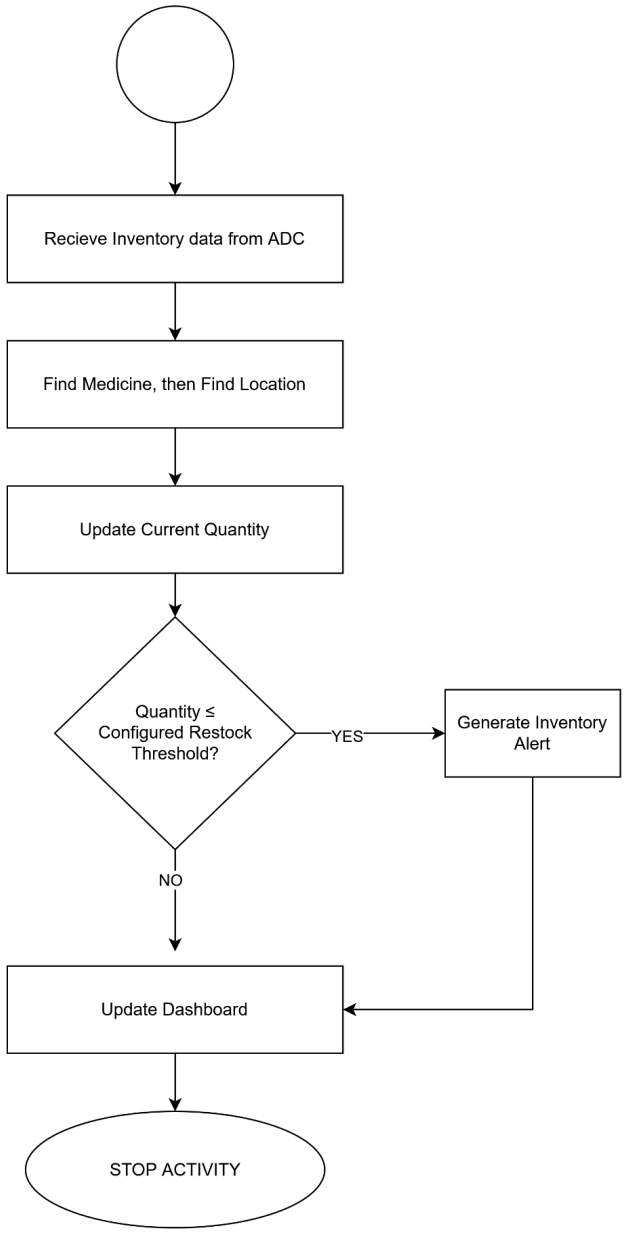
The following use case diagram shows the interactions between the HMS and its primary stakeholders, including pharmacists, nurses, technicians, administrators, and connected to the ADC, which is an actor. The diagram identifies the main functions available to each actor, including inventory supervision, cabinet health monitoring, alarm management, recorded data retrieval, report generation, and dashboard access. The diagram establishes the functional boundaries of the

HMS and demonstrates how stakeholders interact with the system based on their operational responsibilities.

c. Activity diagram(s) (Behaviour):

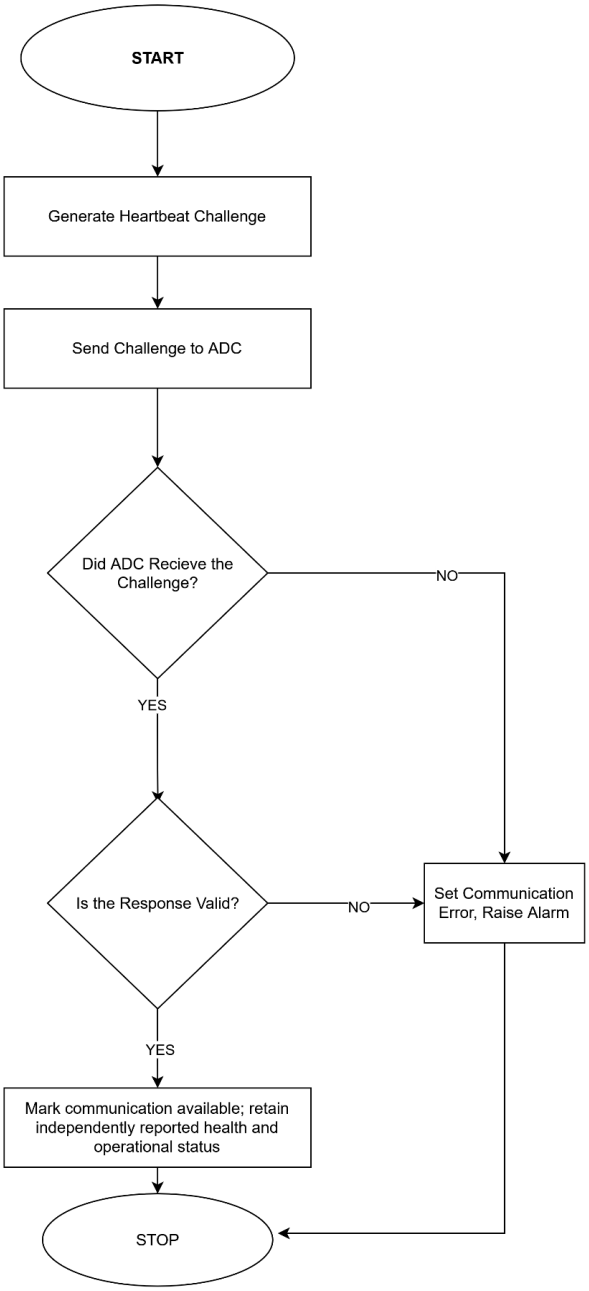
Medication Inventory Monitoring

Activity Diagram #1



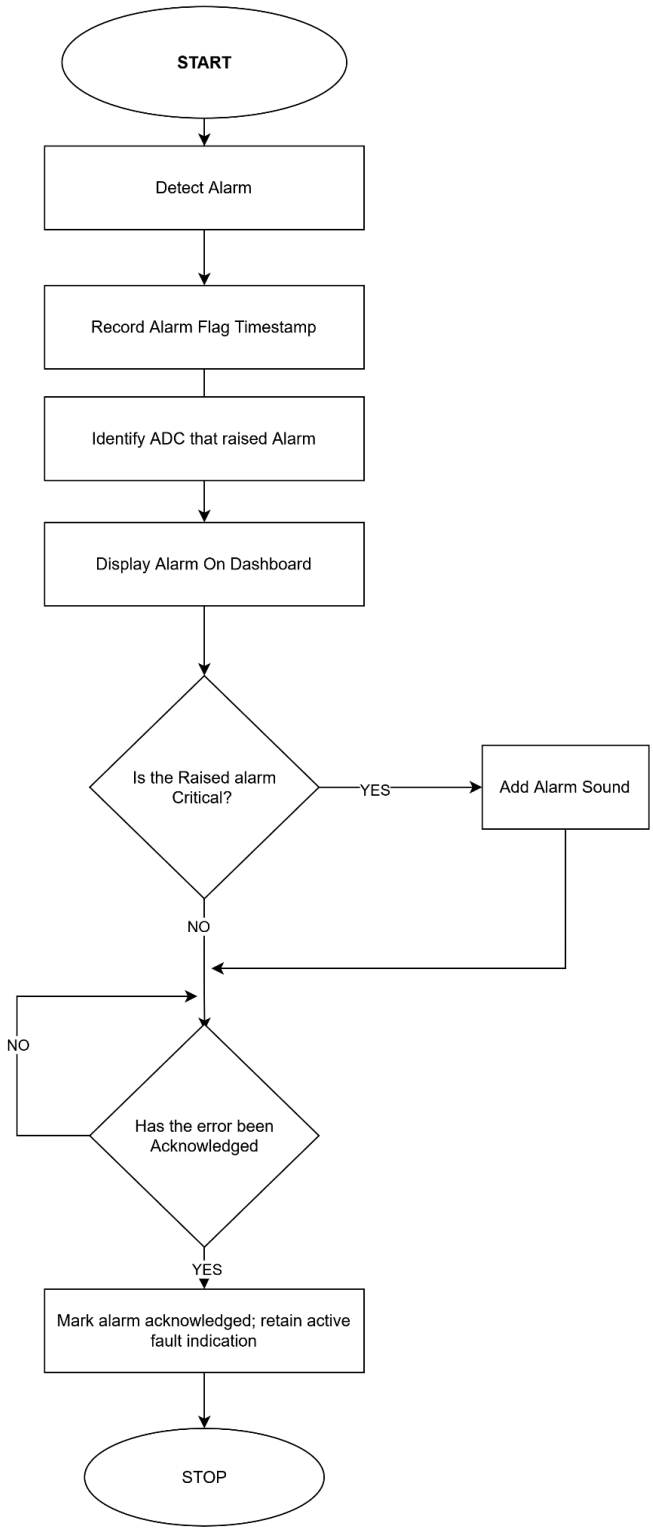
ADC - HMS HEARTBEAT RESPONSE

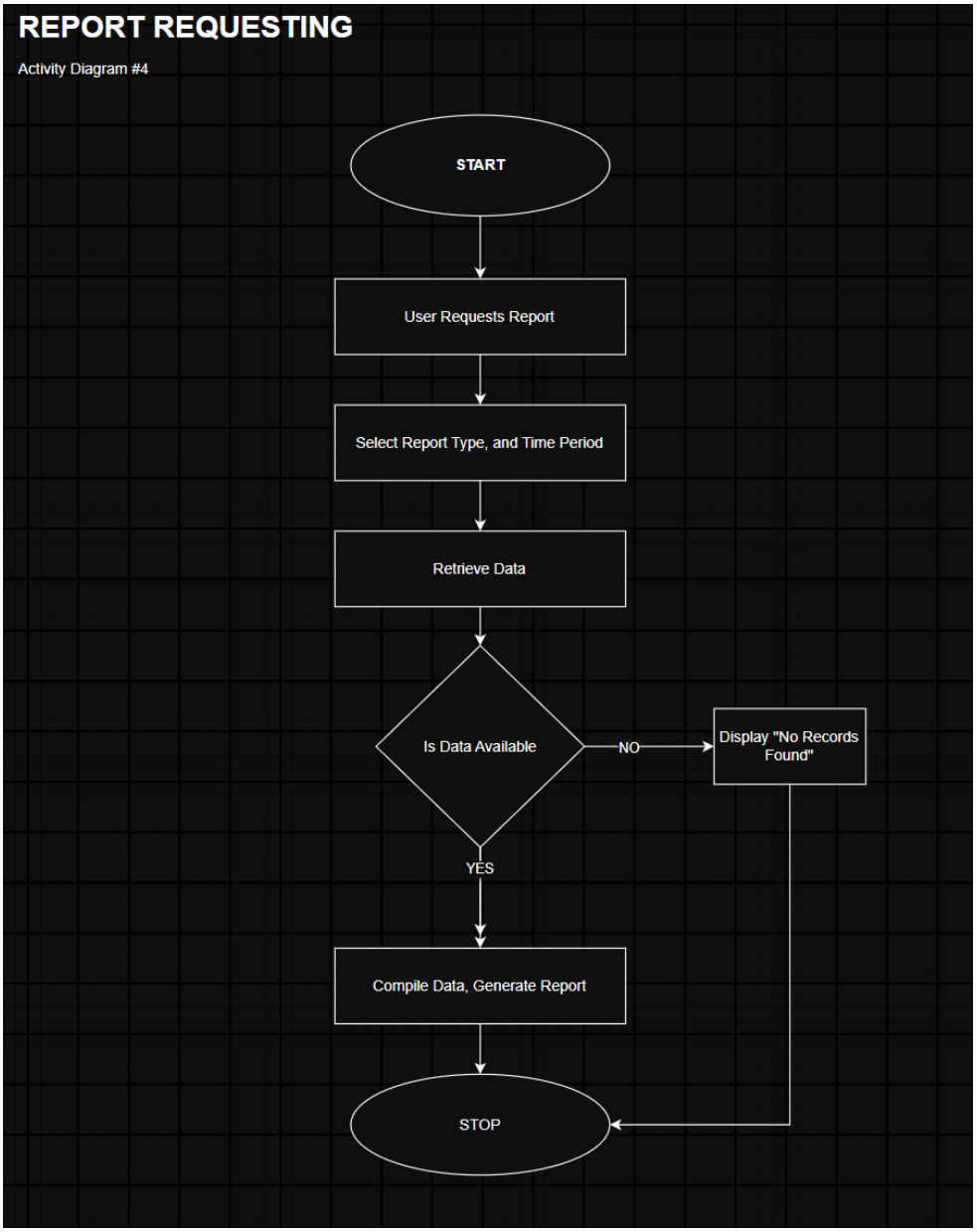
Activity Diagram #2



ALARM HANDLING

Activity Diagram #3

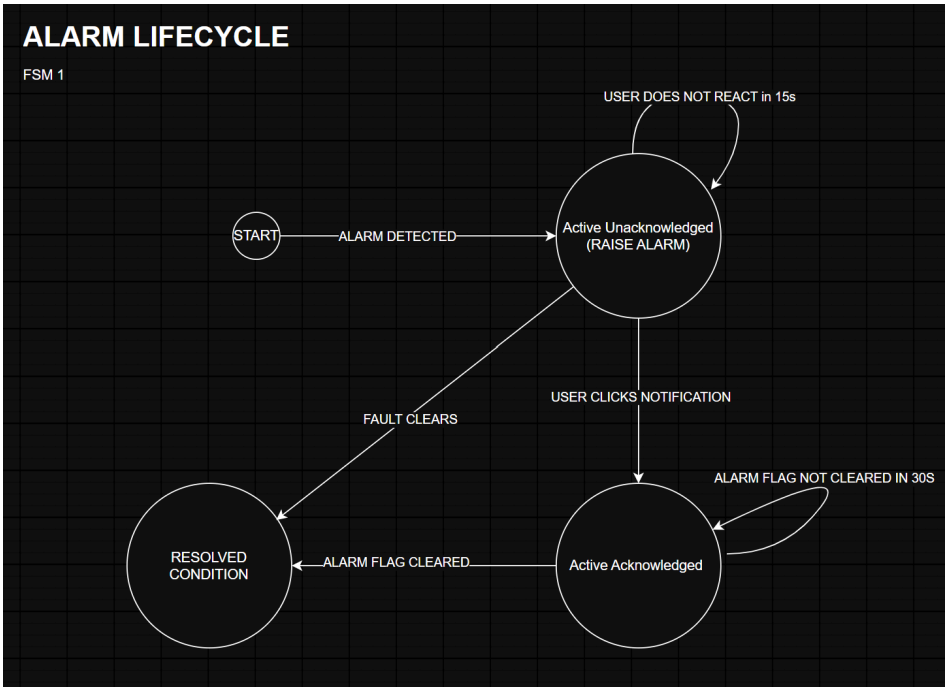


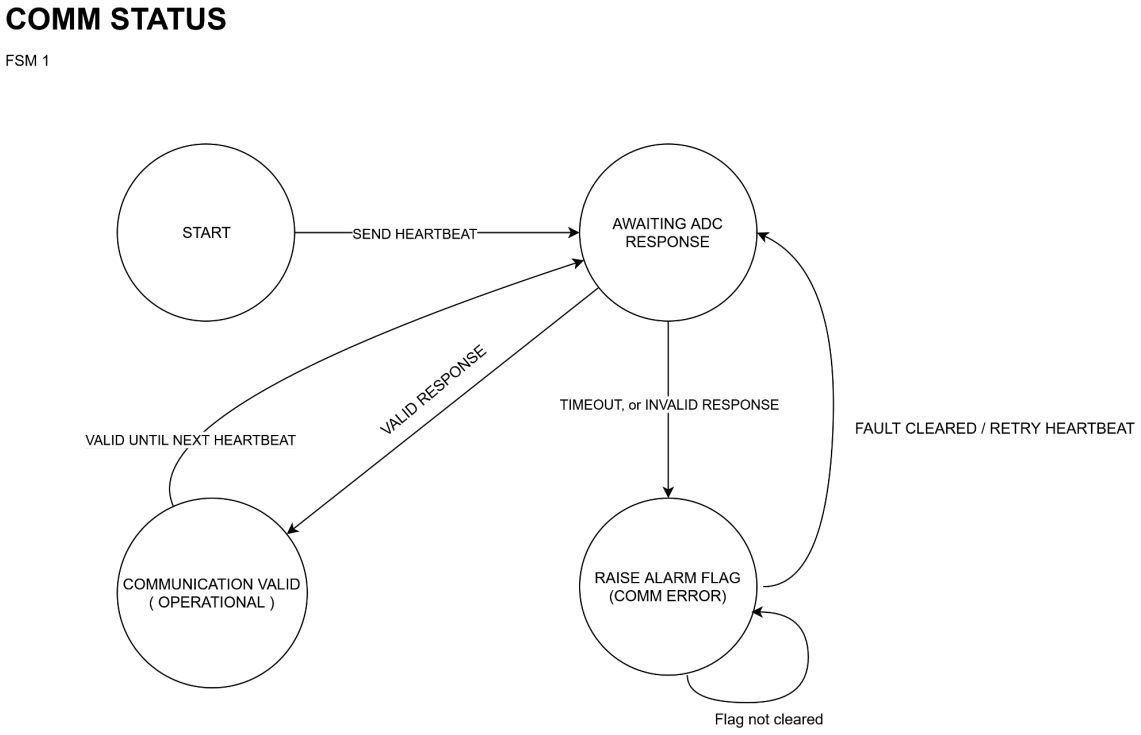


Activity Diagrams

The following activity diagrams illustrate the sequence of operations performed by the HMS when carrying out its primary monitoring and management functions. These diagrams cover medication inventory supervision, ADC heartbeat communication, alarm handling, and report generation. Each activity diagram identifies the actions performed by the system, alongside the decision points that determine how the HMS responds under normal and off-normal operating conditions. This provides a more detailed representation of how information is processed and how the HMS responds to changes in ADC status, inventory information, and user requests.

d. State machine diagram(s) (Behaviour):





State Machine Description

The state machine diagrams depicted below trace the various functional states within the Hospital Monitoring System (HMS), highlighting alarm processing and ADC connectivity. The alarm lifecycle model outlines how an alarm progresses

across active, acknowledged, and resolved statuses driven by physical events and operator interventions. In parallel, the communication model tracks how the HMS validates connection health via regular challenge-response heartbeats, moving through normal operation, response pending, and error states. Combined, these models clarify system behavior across standard and abnormal scenarios, ensuring connection drops and open alerts receive proper tracking.

7. Traceability Table

The following tables connect each Hospital Monitoring System (HMS) requirement to its associated user story or theme, responsible architecture components, and relevant stakeholders or requirement sources.

User-story titles are used as references until unique story identifiers are assigned. Cross-cutting requirements apply across multiple workflows and are identified accordingly.

A. Functional Requirements

Requirement identifier	User story / theme	Architecture component	Stakeholder / source
HMS-ADC-00	ADC Communication	ADC Communication Management	Technician; partner Automated Dispensing Cabinet (ADC) system
HMS-ADC-01	ADC Communication	ADC Communication Management; Configuration Management	Technician; partner ADC system
HMS-ADC-02	ADC Communication	ADC Communication Management	Technician; partner ADC system
HMS-ADC-03	ADC Communication	ADC Communication Management	Partner ADC system; agreed heartbeat mechanism
HMS-ADC-04	ADC Communication	ADC Communication Management	Technician
HMS-ADC-05	ADC Communication	ADC Communication Management	Partner ADC system; Pharmacist; Technician

HMS-ADC-06	ADC Communication / Recorded Data	ADC Communication Management; Audit & Historical Data	Technician; Administrator
HMS-ADC-07	ADC Communication / Alarm Management	ADC Communication Management; Alarm Management	Technician; Pharmacist
HMS-IS-00	Inventory Supervision	Inventory Management	Pharmacist
HMS-IS-01	Inventory Supervision	Inventory Management	Pharmacist
HMS-IS-02	Inventory Supervision	Inventory Management	Pharmacist; partner ADC system
HMS-IS-03	Inventory Supervision	Inventory Management	Pharmacist
HMS-IS-04	Inventory Supervision / Recorded Data	Inventory Management; Audit & Historical Data	Pharmacist
HMS-IS-05	Inventory Supervision	Inventory Management; Graphic User Interface	Pharmacist
HMS-IS-06	Inventory Supervision	Inventory Management; Configuration Management	Pharmacist
HMS-IS-07	Inventory Supervision	Inventory Management; Configuration Management	Pharmacist
HMS-IS-08	Inventory Supervision / Alarm Management	Inventory Management; Configuration Management; Alarm Management	Pharmacist
HMS-IS-09	Inventory Supervision / Alarm Management	Inventory Management; Alarm Management	Pharmacist
HMS-IS-10	Inventory Supervision	Inventory Management; Graphic User Interface	Pharmacist; Nurse
HMS-IS-11	Inventory Supervision	Inventory Management; Graphic User Interface	Authorized inventory operator

HMS-IS-12	Inventory Supervision / Alarm Management	Inventory Management; Alarm Management	Authorized inventory operator; Pharmacist
HMS-AM-00	Alarm Management / Inventory Supervision	Alarm Management; Inventory Management	Pharmacist; partner ADC system
HMS-AM-01	Alarm Management / Cabinet Health Monitoring	Alarm Management; Cabinet Health Monitoring	Technician; Nurse
HMS-AM-02	Alarm Management / ADC Communication	Alarm Management; ADC Communication Management	Technician
HMS-AM-03	Alarm Management; access control across workflows	Alarm Management; User & Access Management	Administrator
HMS-AM-04	Alarm Management / Cabinet Health Monitoring	Alarm Management; Cabinet Health Monitoring	Nurse; Technician; partner ADC system
HMS-AM-05	Alarm Management / ADC Communication	Alarm Management; ADC Communication Management	Pharmacist; Technician
HMS-AM-06	Alarm Management — schedule-conflict scenario required	Alarm Management; Operational Database	Pharmacist; Physician
HMS-AM-07	Alarm Management	Alarm Management; Graphic User Interface	Authorized operator; Technician
HMS-AM-08	Alarm Management / Recorded Data	Alarm Management; Audit & Historical Data	Pharmacist; Administrator
HMS-AM-09	Alarm Management / Dashboard	Alarm Management; Graphic User Interface	Pharmacist; Nurse
HMS-RD-00	Recorded Data	Audit & Historical Data; Operational Database	Administrator; Pharmacist
HMS-RD-01	Recorded Data	Audit & Historical Data; Operational Database	Administrator; Pharmacist

HMS-RD-02	Recorded Data	Audit & Historical Data; Operational Database	Administrator; Pharmacist
HMS-RD-03	Recorded Data / Inventory Supervision	Audit & Historical Data; Inventory Management	Administrator; Pharmacist
HMS-RD-04	Recorded Data / Alarm Management	Audit & Historical Data; Alarm Management	Administrator
HMS-RD-05	Recorded Data / Cabinet Health Monitoring	Audit & Historical Data; Cabinet Health Monitoring	Technician; Administrator
HMS-RD-06	Recorded Data / Cabinet Health Monitoring	Audit & Historical Data; Cabinet Health Monitoring	Technician; Administrator
HMS-RD-07	Recorded Data	Audit & Historical Data; Operational Database	Administrator; Pharmacist
HMS-RD-08	Recorded Data; access control across workflows	Audit & Historical Data; User & Access Management	Administrator
HMS-RD-09	Recorded Data / Inventory Supervision	Audit & Historical Data; Inventory Management	Administrator; Pharmacist
HMS-RG-00	Report Generation	Reporting Module; Audit & Historical Data	Administrator; Pharmacist
HMS-RG-01	Report Generation	Reporting Module; Audit & Historical Data	Administrator; Pharmacist
HMS-RG-02	Report Generation	Reporting Module; Audit & Historical Data	Administrator; Technician
HMS-RG-03	Report Generation	Reporting Module; Audit & Historical Data	Administrator; Pharmacist
HMS-RG-04	Report Generation	Reporting Module; Audit & Historical Data	Administrator
HMS-RG-05	Report Generation	Reporting Module; Audit & Historical Data	Administrator
HMS-RG-06	Report Generation	Reporting Module	Administrator

HMS-DB-00	Dashboard	Graphic User Interface; ADC Communication Management	Pharmacist; Nurse
HMS-DB-01	Dashboard / Inventory Supervision	Graphic User Interface; Inventory Management	Pharmacist; Nurse
HMS-DB-02	Dashboard / Alarm Management	Graphic User Interface; Alarm Management	Pharmacist; Nurse
HMS-DB-03	Dashboard / Alarm Management	Graphic User Interface; Alarm Management	Pharmacist; Nurse
HMS-DB-04	Dashboard / Recorded Data	Graphic User Interface; Audit & Historical Data	Pharmacist; Administrator
HMS-DB-05	Dashboard / ADC Communication / Cabinet Health Monitoring	Graphic User Interface; ADC Communication Management; Cabinet Health Monitoring	Pharmacist; Nurse; Technician
HMS-DB-06	Dashboard / Inventory Supervision	Graphic User Interface; Inventory Management	Pharmacist; Nurse
HMS-CHM-00	Cabinet Health Monitoring / ADC Communication	Cabinet Health Monitoring; ADC Communication Management	Nurse; Technician; partner ADC system
HMS-CHM-01	Cabinet Health Monitoring	Cabinet Health Monitoring	Nurse; Technician
HMS-CHM-02	Cabinet Health Monitoring	Cabinet Health Monitoring	Nurse; Technician
HMS-CHM-03	Cabinet Health Monitoring	Cabinet Health Monitoring	Technician
HMS-CHM-04	Cabinet Health Monitoring / Recorded Data	Cabinet Health Monitoring; Audit & Historical Data	Technician
HMS-CHM-05	Cabinet Health Monitoring	Cabinet Health Monitoring	Nurse; Technician
HMS-CHM-06	Cabinet Health Monitoring	Cabinet Health Monitoring	Nurse; Technician

HMS-CHM-07	Cabinet Health Monitoring / Recorded Data	Cabinet Health Monitoring; Audit & Historical Data	Technician
HMS-CHM-08	Cabinet Health Monitoring / Alarm Management	Cabinet Health Monitoring; Alarm Management	Nurse; Technician
HMS-CHM-09	Cabinet Health Monitoring / ADC Communication	Cabinet Health Monitoring; ADC Communication Management	Nurse; Technician

B. Non-Functional Requirements

Requirement identifier	Requirement summary	User story / theme	Architecture component	Stakeholder / source
SAF-00	Block state-changing requests when ADC communication is unavailable or unverified.	ADC Communication; supervisory-request workflows	ADC Communication Management	Authorized operator; partner ADC system
SAF-01	Identify stale or unavailable information.	Dashboard / ADC Communication / Cabinet Health Monitoring	Graphic User Interface; ADC Communication Management; Cabinet Health Monitoring	Pharmacist; Nurse; Technician
SAF-02	Keep acknowledgement separate from fault resolution and operational status.	Alarm Management / Dashboard	Alarm Management; Cabinet Health Monitoring; Graphic User Interface	Pharmacist; Nurse; Technician
SEC-00	Restrict protected functions and records to authenticated users.	Cross-cutting: all protected workflows	User & Access Management	All authorized users; Administrator

SEC-01	Record authentication failures and denied operations.	Recorded Data; cross-cutting access control	User & Access Management; Audit & Historical Data; Audit Log Database	Administrator
SEC-02	Enforce the documented role-permission matrix.	Cross-cutting: all protected workflows	User & Access Management	All authorized users; Administrator
SEC-03	Authenticate communication endpoints and protect exchanged information.	ADC Communication; cross-cutting security	ADC Communication Management	Administrator; partner ADC system
REL-00	Continue monitoring other ADCs when one disconnects.	ADC Communication / Dashboard	ADC Communication Management; Graphic User Interface	Pharmacist; Nurse; Technician
REL-01	Reject invalid messages without corrupting accepted information or interrupting other monitoring.	ADC Communication / Inventory Supervision / Cabinet Health Monitoring	ADC Communication Management; Inventory Management; Cabinet Health Monitoring	Pharmacist; Technician
REL-02	Preserve saved records and configuration across normal shutdown and restart.	Recorded Data; cross-cutting persistence	Configuration Management; Audit & Historical Data; Operational Database; Audit Log Database	Administrator
REL-03	Prevent duplicate events from duplicating transactions or inventory changes.	ADC Communication / Inventory Supervision / Recorded Data	ADC Communication Management; Inventory Management; Audit & Historical Data	Pharmacist; Administrator

AVA-00	Retain access to historical records during ADC disconnection and label affected live information.	Recorded Data / Dashboard / ADC Communication	Audit & Historical Data; Graphic User Interface; ADC Communication Management	Pharmacist; Nurse; Administrator
AVA-01	Resume monitoring after reconnection and confirm freshness before marking information current.	ADC Communication / Dashboard / Cabinet Health Monitoring	ADC Communication Management; Inventory Management; Cabinet Health Monitoring; Graphic User Interface	Pharmacist; Nurse; Technician
PER-00	Meet the agreed dashboard-query response limit under the documented workload.	Dashboard	Graphic User Interface; HMS Application Services	Pharmacist; Nurse; Administrator
PER-01	Detect an unanswered heartbeat within the configured timeout.	ADC Communication	ADC Communication Management	Technician; partner ADC system
PER-02	Display accepted updates and generated alarms within agreed limits.	Dashboard / Alarm Management	Graphic User Interface; Alarm Management; ADC Communication Management	Pharmacist; Nurse
TRA-00	Retain transactions, alarms, acknowledgements, and threshold changes.	Recorded Data / Alarm Management / Inventory Supervision	Audit & Historical Data; Inventory Management; Alarm Management; Configuration Management	Administrator; Pharmacist

TRA-01	Associate recorded events with identifying information.	Recorded Data	Audit & Historical Data; Audit Log Database	Administrator
TRA-02	Retain event identity and order and identify gaps in historical reconstruction.	Recorded Data / Inventory Supervision	Audit & Historical Data; Inventory Management	Administrator; Pharmacist
USA-00	Use consistent identifiers, labels, and units.	Dashboard / Inventory Supervision / Recorded Data / Report Generation	Graphic User Interface; Inventory Management; Audit & Historical Data; Reporting Module	All authorized users
USA-01	Communicate alarm severity and status without relying solely on colour, blinking, or sound.	Alarm Management / Dashboard	Graphic User Interface; Alarm Management	Pharmacist; Nurse; users with accessibility needs
USA-02	Distinguish communication, health, operational status, and freshness.	Dashboard / ADC Communication / Cabinet Health Monitoring	Graphic User Interface; ADC Communication Management; Cabinet Health Monitoring	Pharmacist; Nurse; Technician
USA-03	Support keyboard operation and visible focus for core workflows.	Inventory Supervision / Alarm Management / Report Generation	Graphic User Interface	Users requiring or preferring keyboard operation

8. Assumptions and decisions

The following assumptions and decisions were established to define the operational scope of the Hospital Monitoring System (HMS) and address areas where specific

implementation details have not been provided. These assumptions support the development of the system requirements, architecture, and UML models, while considering the safety, security, and accessibility of the overall systems.

Assumptions

1. The following assumptions define the operating environment and responsibility boundaries of the Hospital Monitoring System (HMS). They do not replace the functional and non-functional requirements. Assumptions affecting communication or shared responsibilities shall be confirmed with the Automated Dispensing Cabinet (ADC) sub-team.
2. **1. Multiple Cabinets:** The hospital operates multiple ADCs that are monitored through a common HMS. Each cabinet is assigned a unique cabinet_id, which is used to associate messages, inventory locations, alarms, and historical records with the correct device. Cabinets may become unavailable independently; the absence of one cabinet does not imply that all cabinets are unavailable.
3. **2. ADC Communication Interface:** Each ADC provides an interface for exchanging the information required by the HMS, including inventory updates, medication transactions, health information, alarms, and maintenance records. Message formats, required fields, identifiers, units, and response meanings shall be agreed with the ADC sub-team. The HMS accesses cabinet information through this interface rather than through direct access to ADC internals.
4. **3. Hospital Network:** The hospital provides a network capable of connecting the HMS and registered ADCs. The network may experience delays, interruptions, and reconnections; continuous connectivity is not assumed. Network installation and administration are outside the HMS software scope, while detection and handling of communication failures remain HMS responsibilities under the requirements.
5. **4. User Accounts and Roles:** Hospital personnel are provided with valid accounts and assigned roles appropriate to their responsibilities. A documented role-permission matrix determines which operations and records each role may access. Account availability does not imply unrestricted access: the HMS remains responsible for authenticating users and enforcing permissions under SEC-00 and SEC-02.
6. **5. Available ADC Information:** Each ADC supplies the operational information needed for the monitoring capabilities agreed between the sub-teams. Information about optional equipment, such as backup power supplies or active cooling systems, is available only where that equipment is installed and supported by the interface. The absence of an optional feature shall be distinguished from a reported failure or a missing update.

7. **6. Local Hardware and Safety Responsibilities:** Each ADC controls its own dispensing hardware, including locks, sensors, and actuators, and enforces the local safety conditions required for its operation. The HMS provides supervision and exchanges agreed requests and configuration information; it does not directly operate those hardware components. Loss of HMS communication does not transfer local hardware-control responsibility to the HMS.
8. **7. Event Identification and Timing:** ADC messages provide the identifiers and timestamps needed to associate events with the correct cabinet, medication, inventory location, and operator where applicable. The sub-teams shall agree on event-identification, ordering, and timestamp conventions, including any clock-synchronization assumptions. These conventions support duplicate detection, freshness checks, and historical reconstruction; the HMS does not assume that messages always arrive once or in chronological order.
9. **8. Persistent Storage:** A persistent storage facility is available to retain HMS configuration, medication transactions, alarm records, and maintenance history. Its implementation may use a suitable storage mechanism selected during design; these requirements do not mandate a particular database product. Successfully saved information is expected to remain available across normal application shutdown and restart, as required by REL-02.
10. **9. Challenge-Response Heartbeat:** Each ADC supports the mathematical challenge-response heartbeat mechanism defined in HMS-ADC-02–04. The HMS sends challenges at the configured interval, and the ADC returns the corresponding answers. The agreed interface defines permitted challenges, response matching, and timeout behavior. A valid response establishes communication availability but does not independently establish cabinet health or operational readiness.
11. **10. Operator Devices and Accessibility:** Hospital staff access the HMS through devices compatible with its selected user-interface platform. The intended operating environment supports displaying inventory, alarms, reports, and cabinet status and provides keyboard input for the workflows specified in USA-03. Device availability does not replace the HMS obligation to provide clear status labels, visible keyboard focus, and alarm indications that do not rely solely on colour or sound.
12. **11. Reported Information and Physical Accuracy:** ADCs are responsible for measuring physical inventory and hardware conditions and reporting those observations through the agreed interface. The HMS relies on those reports and does not independently measure the cabinet's physical contents. It still validates received messages, identifies stale information, and prevents duplicate events from being applied more than once. Authorized operators may report

discrepancies between physical stock and recorded quantities under HMS-IS-11–12.

13. **12. Deployment Infrastructure and Recovery:** The hospital provides the computing environment needed to operate the HMS. Backup servers, replicated databases, and related redundancy shown in the architecture represent intended deployment infrastructure rather than additional implementation commitments established by this deliverable. The HMS recovery obligations remain those explicitly stated in the requirements, including persistence across normal shutdown and restart, continued monitoring of unaffected cabinets, and resumption of monitoring after valid communication is restored.
14. **13. External Patient and Medication Information:** Where schedule-conflict detection or other specified functions require information from the Hospital Information System or Pharmacy System, that information is supplied through an agreed interface or a documented simulation. The source of each data category, its identifiers, update behavior, and supported conflict rules shall be established before testing the corresponding capability. Access to an external system does not imply responsibility for maintaining that system or independently determining clinical treatment rules.
15.

Design Decisions

1. **Separation of HMS and ADC Responsibilities:** The HMS will operate as a centralized monitoring and management application, while individual ADCs remain responsible for physical dispensing operations. This separation prevents the HMS from directly controlling dispensing hardware.
2. **Centralized Monitoring:** Information received from multiple ADCs will be presented through a centralized dashboard, allowing hospital staff to monitor cabinet health, inventory levels, alarms, and operational status without accessing each ADC individually.
3. **Communication Safety:** The HMS shall prevent state-changing supervisory requests from being sent to an ADC when communication is unavailable or has not been verified, as specified in SAF-00. Heartbeat and connection-recovery exchanges remain available so that communication can be re-established. Communication status shall be evaluated separately from cabinet health and operational status; receiving a valid heartbeat shall not clear an existing hardware or sensor fault.
4. **Role-Based Access Control:** Access to HMS functions and recorded information will be restricted according to assigned user roles, ensuring that users can only access information and perform actions relevant to their responsibilities.

5. **Historical Data and Accountability:** Medication transactions, maintenance activities, alarms, and significant system events will be recorded to support historical review, report generation, and accountability.
6. **Alarm and Fault Identification:** The HMS will identify and display abnormal ADC conditions, allowing hospital staff to recognize cabinets requiring attention and respond accordingly.
7. **Patient Privacy and Data Security:** Patient-related medication records and other sensitive information will only be accessible to authorized users. The system will prioritize the protection of confidential information throughout data storage, communication, and reporting.
8. **Accessibility and Usability:** The HMS dashboard will present operational information using clear labels, distinguishable status indicators, and understandable alarm messages. Important conditions will not rely exclusively on colour to ensure information remains accessible to users with different visual abilities.
9. **System Reliability and Recovery:** The HMS shall continue monitoring other ADCs when one connection fails and shall retain access to previously recorded information while identifying affected live information as stale or unavailable. Following reconnection, information shall be marked current only after a valid fresh update is received. Successfully saved configuration, transaction, and alarm records shall persist across normal application shutdown and restart. These decisions implement REL-00, REL-02, and AVA-00–01; they do not establish an additional guarantee of automatic server failover or recovery from every type of system failure.
10. **Reporting and Inventory Supervision:** The HMS will support the generation of reports based on recorded medication transactions, inventory information, and operational events, allowing pharmacists and other authorized personnel to identify recurring issues and evaluate ADC activity.

9. Citations

[1] A. Zavaleta Bernuy, “Software development,” SFWRENG/MECHTRON 3K04, Week 1 lecture slides, McMaster University, Hamilton, ON, Canada, 2026.

[2] A. Zavaleta Bernuy, “Software development styles,” SFWRENG/MECHTRON 3K04, Week 2 lecture slides, McMaster University, Hamilton, ON, Canada, 2026.

[3] A. Zavaleta Bernuy, “Requirements engineering,” SFWRENG/MECHTRON 3K04, Week 3 lecture slides, McMaster University, Hamilton, ON, Canada, 2026.

[4] A. Zavaleta Bernuy, “Modeling the system,” SFWRENG/MECHTRON 3K04, Week 4 lecture slides, McMaster University, Hamilton, ON, Canada, 2026.

[5] A. Zavaleta Bernuy, “*ReQuest*” *ReQuest*, McMaster University. Accessed: September 28, 2026. [Online]. Available: https://request.cas.mcmaster.ca/

[6] Health Level Seven International, “HL7 Version 2 Product Suite.” Accessed: September 29, 2026. [Online]. Available: https://www.hl7.org/implement/standards/product_brief.cfm?product_id=185

[7] Omnicell, “Advancing NHS pharmacy operations: The technical blueprint of Omnicell automation solutions,” Nov. 21, 2024. Accessed: Oct. 1, 2026. [Online]. Available: https://www.omnicell.co.uk/resource-library/article/advancing-nhs-pharmacy-operations-the-technical-blueprint-of-omnicell-automation-solutions/

[8] OpenAI, “ChatGPT,” [GPT-6 Astra]. Accessed: Oct. 18, 2026. [Online]. Available: https://chatgpt.com/